Transformations, Groups and Geometry

UNSW MATH3511 25T3

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Contents

0 Revision

Here is a list of concepts you should know from high school.

- Vertically opposite, alternating, corresponding and co-interior angles.
- Interior angles of a triangle sum to π; exterior angle equals sum of opposite interior angles.
- Congruent and similar triangles.
- Isosceles and equilateral triangles.
- Sine rule: $\frac{a}{\sin\alpha} = \frac{b}{\sin\beta} = \frac{c}{\sin\gamma}$.
- Cosine rule: $a^2 = b^2 + c^2 - 2bc\cos\alpha$. Pythagoras' theorem if $\alpha = \frac{\pi}{2}$.
- Three non collinear points defines a unique circle.
- A quadrilateral is cyclic if its vertices lie on a circle.

1 Triangle Geometry

1.1 Signed Lengths and Orientation

A line through two points A and B is denoted AB, typically *in that direction*. For a point P on the line AB, if P lies between A and B then the ratio $\frac{AP}{PB} > 0$. But if P is outside A and B then $\frac{AP}{PB} < 0$. If P and A are the same point, then the ratio is zero; if P and B are the same, then the ratio is ∞.

An **altitude** of a triangle is a line segment through a given vertex and perpendicular to the line containing the side opposite the vertex.

In a triangle, the orientation is called **positive** if the vertices are taken anti-clockwise in the order given. For an oriented triangle, we take directed lengths to match the orientation: AB, BC and CA are positive in $\triangle ABC$. We extend this convention to area and angle:

- The area of a triangle is positive if the triangle is positively oriented; otherwise it is negative.
- Positive angles go anticlockwise. If edges are signed, we need to account for that. We take $\angle ABC$ as positive if the sense runs anticlockwise from BA to BC when both are the same sign; if they are opposite signs, we change the sign of the angle to compensate.

We denote the sign of the area as $\triangle ABC = \frac{1}{2}|AB| \cdot |BC| \cdot \sin ABC$.

1.2 Intersections

A **centroid** is the intersection of the medians while a **orthocetnre** is the intersection of altitudes. It can be shown that the medians are concurrent. The medians and altitudes are examples of **cevians**.

> **Definition. (Cevian)** In any triangle a line segment joining a vertex to a non-vertex point on the opposite side, possibly extended, is called a cevian.

Theorem. (Ceva's Theorem (1678)) In $\triangle ABC$ the cevians AD, BE and CF (where D is on BC, E is on AC and F is on AB) are concurrent if and only if

$$\frac{AF}{FB} \cdot \frac{BD}{DC} \cdot \frac{CE}{EA} = 1.$$

Corollary. In $\triangle ABC$ the cevians AD, BE and CF are concurrent if and only if

$$\frac{\sin ACF}{\sin FCB} \cdot \frac{\sin BAD}{\sin DAC} \cdot \frac{\sin CBE}{\sin EBA} = 1,$$

taking proper account of orientations of the angles.

1.3 Collinearity

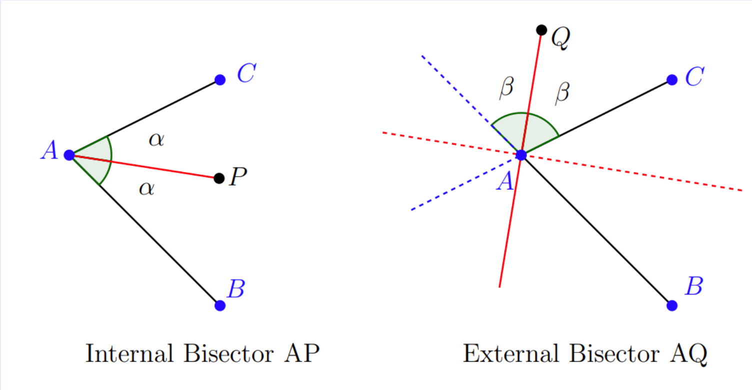
Theorem. (Menelaus' Theorem) In $\triangle ABC$ let D, E and F be on side BC, CA, AB (or their extensions) respectively. Then D, E and F are collienar if and only if

$$\frac{AF}{FB} \cdot \frac{BD}{DC} \cdot \frac{CE}{EA} = -1.$$

Theorem. (Pappus' Theorem) Suppose points A, C, E lie on one line and B, D, F on a second line, where AB, CD and EF form a triangle. Let $L = AB \cap DE$, $M = CD \cap FA$ and $N = EF \cap BC$. Then L, M and N are collinear.

1.4 Angle Bisectors

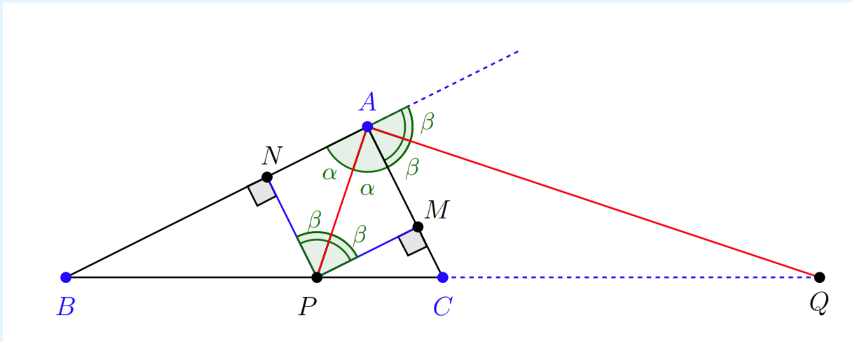
Definition. (Bisectors) In the image below, the line AP in the left hand diagram is the **internal bisector** of $\angle BAC : \angle BAP = \frac{1}{2}\angle BAC$ and the line AQ in the right hand diagram is the **external bisector** of $\angle BAC : \angle CAQ = \frac{1}{2}(\pi - \angle BAC)$, that is it bisects angle between $AC$and the extension of the line AB.



Theorem. (Angle Bisector Theorem) Consider $\triangle ABC$, and its angle bisector as below.

a) If the interior bisector of $\angle BAC$ meets BC at P then $\frac{BP}{PC} = \frac{AB}{CA}$.

b) If the exterior bisector of $\angle BAC$ meets BCat Qthen $\frac{BQ}{QC} = -\frac{AB}{CA}$.



Theorem. (Incentre Theorem) The internal bisectors of a triangle are concurrent.

The point of concurrence is called the **increntre**. This point is equidistance from the three sides, so a circle centred here and tangent to the edges is the largest circle that can fit inside the triangle. This circle is called the **incircle**.

Theorem. (Excentre Theorem) The external bisectors of any two angles of a triangle are concurrent with the internal bisector of the third angle.

The points of concurrence (there are three of them) are called the **excentres**. Again, each excentre is the same distance from the three sides, and so circles centres at the excentres with that distance as radius (the **excircles**) are tangent to the sides of the triangle, one internally, two externally.

Theorem. In $\triangle ABC$, let the internal bisectors at angles A, B and C meet the opposite sides at P, Q and R respectively and the external bisectors meet the opposite sides at P', Q' and R' respectively. Then $\{P, Q, R'\}, \{P, Q', R\}$ and $\{P', Q, R\}$ are triples of collinear points.

1.5 Area Formulas

Let r be the radius of the incircle. Then

$$\triangle ABC = \frac{1}{2}r(a+b+c) = sr,$$

where $s = \frac{1}{2}(a+b+c)$ is the semiperimeter.

We can get similar results from the excircles. Let r_a be the radius of the excircle centred at J_a. Then

$$\triangle BCJ_a = \frac{1}{2}r_a \cdot a, \quad \triangle ABJ_a = \frac{1}{2}r_a \cdot c, \quad \triangle ACJ_a = \frac{1}{2}r_a \cdot b,$$

since r_a is the distance to each edge. Thus,

$$\triangle ABC = \frac{1}{2}r_a(b-a+c) = (s-a)r_a.$$

Then similarly, we have

$$\triangle ABC = (s-b)r_b = (s-c)r_c.$$

After some trigonometry and rearranging to solve for r, we get

$$r = \sqrt{\frac{(s-a)(s-b)(s-c)}{s}}.$$

Then from the previous expression we get **Heron's formula**:

$$\triangle ABC = \sqrt{s(s-a)(s-b)(s-c)}.$$

Then we get the following results,

$$r_a = \sqrt{\frac{s(s-b)(s-c)}{s-a}} \qquad \text{and} \qquad \frac{1}{r} = \frac{1}{r_a} + \frac{1}{r_b} + \frac{1}{r_c}.$$

1.6 Circles Associated to Triangles

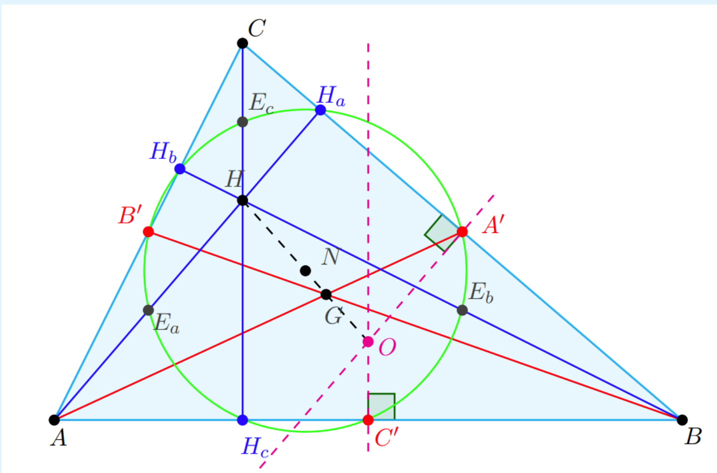
The **circumcircle** goes through the three vertices of a triangle and has centre the **circumcentre**, the intersection of the three perpendicular bisectors of the sides.

Suppose for a circle $\triangle ABC$ with circumcentre O. Let $OA = OB = OC = R$ be the **circumradius**. Then $\angle AOB = 2\angle C$ and so $\angle C = \angle FOB$. Thus $R \sin C = FB = \frac{c}{2}$. Then similar results for the other sides gives the **extended sine rule**:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R.$$

Lemma. The circumcentre of a triangle is the orthocentre of its median triangle.

Theorem. (Nine Point Circle) For $\triangle ABC$, let A', B' and C' be the midpoints of the three sides and H_a, H_b and H_c be the feet of the altitudes. Let H be the orthocentre and E_a, E_b and E_c be the midpoints of the lines from H to the three vertices. Also let O be the circumcentre and R the circumradius. Then if N is the midpoint of the line from H to O, the nine points $A', B', C', H_a, H_b, H_c, E_a, E_b$ and E_c lie on the circle centre N of radius $\frac{R}{2}$.



Theorem. (Feuerbach's Theorem) The nine-point circle is tangent to the incircle and all three excircles.

The point where the incircle and the nine-point circle meet is called the **Feuerbach point**. It is collinear with the incentre and the nine-point centre.

Theorem. (The Euler Line) In any triangle, the orthocentre H, nine-point centre N, circumcentre O and centroid G are collinear. Furthermore N is the midpoint OH and $OG : GH = 1 : 2$.

The line these points lie on is called the **Euler Line**.

Theorem. (Simson Line) Let P be any point on the circumcirle of $\triangle ABC$. Then the feet of the perpendiculars from P to each side of $\triangle ABC$ are collinear.

Proposition. The three Euler points uniquely define a triangle.

1.7 The Cross Ratio

Theorem. Let $\triangle ABC$ have points D, E, F on the sides opposite A, B, C respectively and suppose the cevians AD, BE and CF are concurrent at a point P inside $\triangle ABC$. Let Q be a point on BC.

Then F, E and Q are collinear if and only if $\dfrac{BD \cdot CQ}{CD \cdot BQ} = -1$.

Definition. (Cross Ratio) Let A, B, C, D be any four points. The cross ratio $(A, B; C, D)$ is defined by

$$(A, B; C, D) = \frac{AC}{BC} \div \frac{AD}{BD} = \frac{AC \cdot BD}{BC \cdot AD}.$$

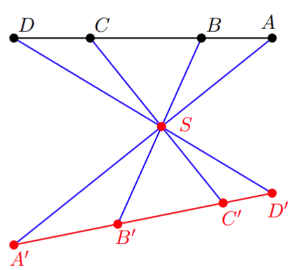
Note that changing the order of the points in $(A, B; C, D)$ may change the value of the cross ratio. However

$$(A, B; C, D) = (B, A; D, C) = (C, D; A, B) = (D, C; B, A),$$

so there are at most 6 different values of the cross ratio.

Definition. (Harmonic) A set of collinear points $\{A, B, C, D\}$ for which $(A, B; C, D) = -1$ are said to be harmonic, and C, D are called the harmonic conjugates of A, B.

Theorem. In the diagram below, we have



$$(A, B; C, D) = (A', B'; C', D') = \left(\frac{\sin \angle ASC}{\sin \angle BSC}\right) \cdot \left(\frac{\sin \angle BSD}{\sin \angle ASD}\right).$$

The sets $\{A, B, C, D\}$ and $\{A', B', C', D'\}$ are said to be in perspective from S, or one set if projected from S. Our result can then be stated as the cross ratio of collinear points is a projective invariant.

Theorem. Suppose that in $\triangle ABC, X$ and Y are points on AB such that CX is the internal bisector of $\angle ACB$ and CY its external bisector. Then $(A, B; X, Y) = -1$, i.e. the four points are harmonic.

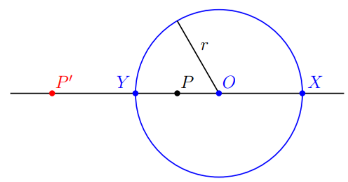
Conversely, given points X and Y on side AB of $\triangle ABC$ with CX orthogonal to CY, if $(A, B; X, Y) = -1$ then CX is the internal and CY the external bisector of $\angle ACB$.

2 Circle Geometry

A circle with centre at a point O and with radius r will be denoted $\mathcal{C}(O, r)$.

2.1 Inversion

Theorem. For circle $\mathcal{C}(O, r)$, let $P \neq O$ be any point and X, Y the intersection of $\mathcal{C}$ and the line through O and P as below. Then $(X, Y; P, P') = -1$ if and only if $OP \cdot OP' = r^2$.



Definition. (Inverse) Let $\mathcal{C}(O, r)$ be a circle. Given a point P distinct from O we can define a point P' called its inverse in $\mathcal{C}$ as follows: on the line OP the point P' is such that $OP \cdot OP' = r^2$.

Definition. The map $T : \mathbb{R}^2 \setminus \{0\} \to \mathbb{R}^2 \setminus \{0\}$ sending each P to its inverse in $\mathcal{C}$ is called inversion. We call O the centre of inversion and r the radius of inversion.

Simple properties of an inversion, T, are immediate:

- T pointwise fixes the circle (if P is on the circle $T(P) = P$);
- T maps the inside of the circle to the outside and vice versa;
- T is an involution, that is $T^2(P) = P$.

Theorem. If a circle $\mathcal{C}(O, r)$ inverts A, B to A', B' respectively, then

$$A'B' = \frac{r^2 AB}{OA \cdot OB}.$$

Theorem. Let A, B, C, D be 4 points all distinct from O. If circle $\mathcal{C}(O, r)$ inverts A, B, C, D to A', B', C', D' respectively, then

$$(A, B; C, D) = (A', B'; C', D').$$

2.2 Poles and Polar

Definition. (Polar and Poles) Let $\mathcal{C}(O, r)$ be a circle. For a point $P \neq O$ its polar, p, is the line through its inverse point P' orthogonal to OP'.

Conversely, given a line ℓ not through O its pole, L, is the inverse point to the point L' closest to O.

Theorem. If P is external to $\mathcal{C}(O, r)$, then its polar, p, is the line through the points where the tangents from Pmeet the circle.

Corollary. There is a ruler and compass construction of the tangents to a circle from an arbitrary external point.

Theorem. For $\mathcal{C}(O, r)$ and a point P that is neither O nor a point on the circle, suppose a line ℓ through P intersects $\mathcal{C}$ at distinct points X and Y.

Then if E is the intersection of ℓ and the polar of P, $\ell$is divided harmonically by X, Y, E and P: i.e. $(X, Y; E, P) = -1$.

Conversely, if E on ℓ is such that $(X, Y; E, P) = -1$, then the line perpendicular to OP through E is the polar of P.

Theorem. Let $\mathcal{C}(O,r)$ be a circle. If ℓ is any line not passing through O, its inverse is a circle through O (minus the point O).

Conversely, if $\mathcal{C}'$ is a circle through O, its inverse is a line perpendicular to the diameter of $\mathcal{C}'$ through O.

Corollary. A circle $\mathcal{C}'$ passing through O is inverted in circle $\mathcal{C}(O,r)$ to a line parallel to the tangent to $\mathcal{C}'$ at O.

Theorem. For $\mathcal{C}(O,r)$ and $L \neq O$, let ℓ be its polar. Then for any point Q on ℓ, the polar of Q, q, passes through L.

Conversely, for ℓ a line not passing through O with pole L, any line q through L has pole Q on ℓ.

Corollary. For given circle $\mathcal{C}(O,r)$, let (P,p) and (Q,q) be two pole/polar pairs, where $N = p \cap q$ exists. Then the line PQ is the polar, n, of N and the pole of PQ is N.

2.3 Intersection of Circles

In general, two curves intersecting at a point P are said to be **orthogonal** if their tangents at P are orthogonal.

Theorem. Two intersecting circles are orthogonal if and only if the angles in opposite segments are complementary.

Theorem. Let A, B, C, D be a harmonic set, $(A,B;C,D) = -1$. The two circles with diameters AB and CD are orthogonal.

Conversely, if two circles with centres O_1 and O_2 are orthogonal, and the line O_1O_2 intersects one circle at A and B and the other at C and D then $(A,B;C,D) = -1$.

2.4 Inverting Circles and Lines

Given a circle $\mathcal{C}(O,r)$, we saw in a previous result t hat a circle through O is inverted to a line not through O, and conversely. It is clear that a line through O is inverted to a line through O, namely the same line.

Theorem. Let $\mathcal{C}(O,r)$ be a circle and $\Sigma(\Omega,\rho)$ a second circle not passing through O. Under inversion in $\mathcal{C}$, Σ is inverted to another circle not through O.

Theorem. Let $\mathcal{C}(O,r)$ be a circle and $\Sigma(\Omega,\rho)$ a second circle meeting $\mathcal{C}$ orthogonally. Then Σ is inverted to itself in $\mathcal{C}$.

When inverting in a circle $\mathcal{C}(O,r)$, we avoided inverting O itself. However, we can imagine adding a **ideal point**, ∞ "at infinity" to the plane, to which O is mapped by inversion in a circle centred at O.

This extended plane has many nice properties, which will explore later, but here we note that we can think of a line in the plane as a circle through ∞ in the extended plane. Then we can sum up our results on inversion as:

Theorem. In the extended plane, inversion in a fixed circle maps a circle to a circle.

Clearly any translation or rotation about a fixed point will send a line to a line and a circle to a circle. It follows that the most general transformation that maps a circle or line to a circle or line is in fact a composition for a translation, a rotation and an inversion.

Thinking of $\mathbb{R}^2$ as $\mathbb{C}$, inversion in the circle $|z-c|=r$ is the map $z \mapsto c+\frac{r^2}{|z-c|^2}(z-c)$ and is the composition of conjugation and a special case of a **linear fractional transformation**,

$$z \mapsto \frac{az+b}{cz+d}.$$

3 Transformation Geometry

3.1 Symmetries and Groups

Definition. (Transformation) A continuous bijection $T : \mathbb{R}^2 \to \mathbb{R}^2$ with a continuous inverse is called a (plane)-transformation.

Definition. (Symmetries) If Γ is a geometric object in the plane, the transformations that preserve (key aspects of) Γ are called symmetrics of Γ.

Definition. (Groups) A group is a non-empty set G on which a binary operation $\star$ is defined, such that the following properites hold:

a) Closure: if $a, b \in G$ then $a \star b \in G$.

b) Associativity: if $a, b, c \in G$ then $(a \star b) \star c) = a \star (b \star c)$.

c) Identity element: there is an element e of G such that for all $a \in G$ we have $a \star e = e \star a = a$

d) Inverses: for each a in G there is an element b of G such that $a \star b = b \star a = e$. This element is usually denoted a^{-1}.

Definition. (Abelian Groups) A group G is abelian if the operation is commutative: $a \star b = b \star a$ for all $a, b \in G$.

It is easy to see that the set of all transformations of the plane is a group under composition.

a) If $T, S : \mathbb{R}^2 \to \mathbb{R}^2$ are continuous bijections, so is $T \circ S$, and it has continuous inverse $S^{-1} \circ T^{-1}$.

b) If $T, S, U : \mathbb{R}^2 \to \mathbb{R}^2$ are any maps on $\mathbb{R}^2$, $(T \circ S) \circ U = T \circ (S \circ U)$.

c) The identity map is a transformation.

d) We have built into the definition the fact that the inverse of a transformation is also a transformation.

Notation: $\mathcal{G}_2$ represents the group of all transformations in the plane.

Definition. (Subgroup) Let G be a group, and let H be a subset of G which is itself a group (under the same operation as G). Then we say that H is a subgroup of G.

Lemma. (The Subgroup Lemma) Let G be a group and H a non-empty subset of G. Then $H$is a subgroup of G if and only if it is closed under the group operation and inverse.

Corollary. The set of symmetries of a geometric figure is a group.

3.2 Affine Transformations

Definition. (Affine Transformation) A transformation $T \in \mathcal{G}_2$ is an affine transformation or an affinity if it maps lines to lines.

Obvious examples of affinities are translations $\mathbf{x} \to \mathbf{x} + \mathbf{b}$ for fixed $\mathbf{b}$, and reflections in a line.

Theorem. The set of all affinities in the plane, $\mathcal{A}_2$, is a group under composition.

Theorem. A transformation $T \in \mathcal{A}_2$ if and only if it is of the form

$$T(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$$

for a fixed invertible matrix A and constant vector $\mathbf{b}$.

Theorem. Let T be an affinity.

a) T will preserve collinearity, concurrence, parallelism, betweenness, line segments, polygons and the ratio of distances on the same or parallel lines.

b) T will in general not preserve angles, areas or circularity.

Proof.

- For collinearity, suppose A, B and C are collinear points and ℓ the line through them. Then $T(\ell)$ is also a line by definition and clearly it goes through $T(A), T(B)$ and $T(C)$.
- For concurrence, if ℓ and λ are two lines and $P = \ell \cap \lambda$ then $T(P) = T(\ell) \cap T(\lambda)$.
- For parallelism, if ℓ and λ are parallel lines but $P = T(\ell) \cap T(\lambda)$ is not empty, then $T^{-1}(P) \in \ell \cap \lambda = \emptyset$, contradiction.
- Let $T(\mathbf{x}) + Ax + \mathbf{b}$.
  - Suppose R lies between P and Q, so that $\mathbf{r} = \mathbf{p} + \lambda(\mathbf{q} - \mathbf{p})$ with $\lambda \in (0, 1)$. Then $T(\mathbf{r}) = A\mathbf{p} + \mathbf{b} + \lambda(A\mathbf{q} - A\mathbf{p})$, and so $T(R)$ is between $T(P) = A\mathbf{p} + \mathbf{b}$ and $T(Q) = A\mathbf{q} + \mathbf{b}$.
- It follows from the previous part that T preserves line segments and hence polygons.
- Since we now know T preserves the order of points on a line, we only need to check the ratio of undirected distances.
  - Suppose P and Q lie on a line ℓ which we write as $\mathbf{a} + \lambda\mathbf{c}$. Let $\mathbf{p} + \mathbf{a} + t\mathbf{c}$ and $\mathbf{q} + \mathbf{a} + s\mathbf{c}$, so $|PQ| = |t - s| \cdot \|\mathbf{c}\|$. If $P' = T(P)$ and $Q' = T(Q)$, then $\mathbf{p}' = A\mathbf{a} + \mathbf{b} + tA\mathbf{c}$ and $\mathbf{q}' = A\mathbf{a} + \mathbf{b} + sA\mathbf{c}$, so that $|P'Q'| = |t - s| \cdot \|A\mathbf{c}\|$.
  - Likewise for a pair U, V on ℓ or a line parallel to ℓ, we might get $|UV = |\tau - \sigma| \cdot \|\mathbf{c}\|$ and $|U'V'| = |\tau - \sigma| \cdot \|A\mathbf{c}\|$.

– So
$$\frac{|PQ|}{|UV|} = \frac{|t-s|}{|\tau-\sigma|} = \frac{|P'Q'|}{|U'V'|},$$
and the ratio of distances is preserved.

- To prove that T may not preserve angles, areas and circularity, we need only consider an example, such as $T(\mathbf{x}) = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} \mathbf{x}$.

Clearly then, an affinity can be thought of as a linear map composed with a translations which is continuous and differentiable. In fact, as an affinity is invertible it is a change of coordinates. From several variable calculus the Jacobian of that coordinate change gives us how transform.

It is easy to check that the Jacobian of $\mathbf{x} \mapsto A\mathbf{x} + \mathbf{b}$ is simply $|\det(A)|$, and so we have proved:

Theorem. An affinity $T(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$ multiplies areas of geometric figures by $|\det(A)|$.

Definition. (Equiareal) A transformation that preserves area is called equiareal.

Clearly, equiareal affinities have $\det(A) = \pm 1$.

Definition. (Positive Angle) The angle between the lines AB and AC taken in that order and the implied direction is positive if
$$\det(\mathbf{x} \mid \mathbf{y}) = \det \begin{pmatrix} x_1 & y_1 \\ x_2 & y_2 \end{pmatrix} > 0 \text{ for } \overrightarrow{AB} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \overrightarrow{AC} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}.$$

Under the affinity $T(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$, the determinant used to define the orientation of the angle will be simply multiplied by $\det(A)$.

So if $\det(A) > 0$ the orientation is preserved: we call such affinities **direct**. If the orientation of angles is reversed the affinity is **indirect**.

3.2.1 Applications of Affinities

Theorem. Let ℓ_1 and ℓ_2 be two lines in $\mathbb{R}^n$. There is an affinity T such that $T(\ell_1) = \ell_2$.

Proof. Write the equations of the lines as $\mathbf{x} + \lambda\mathbf{p} + \mathbf{b}$ and $fx = \lambda\mathbf{q} + \mathbf{c}$ for suitable vectors $\mathbf{p}, \mathbf{q}, \mathbf{b}$ and $\mathbf{c}$. We can easily find an invertible matrix A such that $A\mathbf{p} = \mathbf{q}$, and then the affinity
$$T(\mathbf{x}) = A\mathbf{x} + \mathbf{c} - A\mathbf{b}$$
maps the first line to the second.

Theorem. For any two sets of three non-collinear points $\{P, Q, R\}$ and $\{P', Q', R'\}$ in $\mathbb{R}^2$ there is a unique affinity T such that $T(P) = P', T(Q) = Q'$ and $T(R) = R'$.

Then to summarise the previous two results we say that *all lines* and *all triangles* are **affinely equivalent**.

Theorem. A non-trivial conic section is affinely equivalent to one of the following **standard forms**:

i) $x^2 + y^2 = 1$ for an ellipse;

ii) $x^2 - y^2 = 1$ for a hyperbola;

iii) $y = x^2$ (or, equivalently, $x = y^2$) for a parabola;

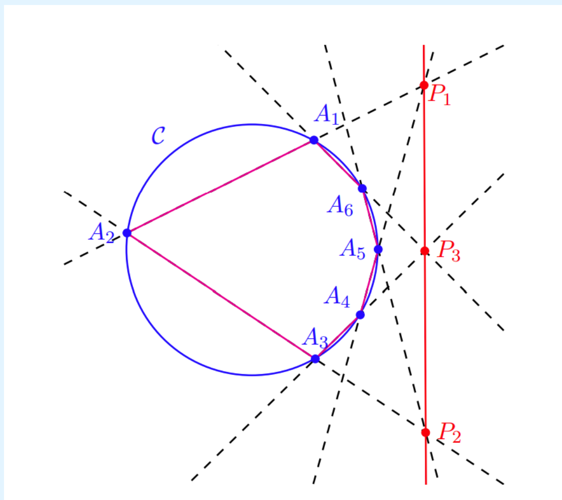
iv) $x^2 = y^2$ for intersecting lines;

v) $x^2 = 1$ (or, equivalently, $y^2 = 1$) for parallel lines;

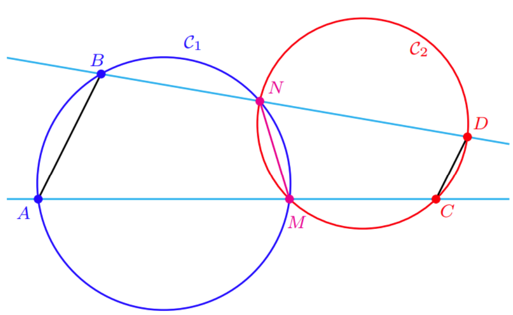
vi) $x = 0$ for one line.

Furthermore, the coordinates in which the conic takes its standard form are orthogonal.

Theorem. (Pascal's Theorem (1639)) Let A_1, A_2, A_3, A_4, A_5 and A_6 be six points on an ellpise $\mathcal{C}_1$. let $P_1 = A_1A_2 \cap A_4A_5, P_2 = A_2A_3 \cap A_5A_6$ and $P_3 = A_3A_4 \cap A_6A_1$be the points where opposite sides of the hexagon $A_1A_2A_3A_4A_5A_6$ intersect. Then (assuming all three points exists) P_1, P_2 and P_3 are collinear.



Lemma. Let $\mathcal{C}_1$ and $\mathcal{C}_2$ be two circles that intersect at M and N. Let AB be any chord of $\mathcal{C}_1$ and C, D be the second intersections of $AM$and BN respectively with $\mathcal{C}_2$, as shown. Then AB and CD are parallel.



3.3 Isometries

Let $d : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$ be the usual (Euclidean) distance. If $A = (a, b)$ and $B = (c, d)$ in some Cartesian coordinate system,

$$d(A, B) = \sqrt{(a - c)^2 + (b - d)^2}.$$

Definition. (Isometry) An isometry, T, is a transformation of the plane that preserves distance:

$$d(T(A), T(B)) = d(A, B) \text{ for all } A, B \in \mathbb{R}^2.$$

Definition. (Congruent) Two figures F_1, F_2 in the plane are congruent if there is an isometry mapping one to the other.

Theorem. The set of isometries under composition is a subgroup of $\mathcal{A}_2$ and is denoted $\mathcal{I}_2$.

Theorem. An isometry T preserves betweenness, collinearity, parallelism, line segments, triangles, and polygons. It also preserves

i) the standard dot product on $\mathbb{R}^2$;

ii) (sizes of) angles;

iii) circles and conic sections.

Proof. The first of properties follows because an isometry is an affinity, but could be easily proved directly.

i) Use the polarisation identity $4(\mathbf{x} \cdot \mathbf{y}) = \|\mathbf{x} + \mathbf{y}\|^2 - \|\mathbf{x} - \mathbf{y}\|^2$.

ii) This follows from (i).

iii) Circles are loci a fixed distance from some central point O, and conic sections are loci of points P such that for some point F (the focus), the ratio PF/d is constant, where d is the distance of P from some fixed line (the directrix).

Theorem. An affinity T of the form $T(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$ is an isometry if and only if A is orthogonal, i.e. $A^T A = I$.

3.3.1 Direct Isometries

Definition. (Direct Isometries) An isometry is direct if it preserves orientation of angles. An isometry that reverses the orientation of angles is called indirect.

Proposition. The set of direct isometries form a group.

The set of indirect isometries is not a group, as it is not closed under composition. Additionally, it does not contain the identity.

Definition. (Translation) A translation $T_{\mathbf{a}} : \mathbb{R}^2 \to \mathbb{R}^2$ is a map defined by $T_{\mathbf{a}}(\mathbf{x}) = \mathbf{x} + \mathbf{a}$.

Theorem. The set of translations is an abelian group of direct isometries under composition.

Definition. (Rotation) A rotation through angle θ about the origin of coordinates is given by

$$U_\theta(\mathbf{x}) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \mathbf{x}$$

in cartesian coordinates. It is thus a linear map on $\mathbb{R}^2$.

Theorem. Both the set of translations and set of rotation about the origin form abelian subgroups of direct isometries under composition.

Definition. A rotation through angle θ about a point P will be denoted $R_{\theta,P}$.

Theorem. A rotation through angle θ about a point P is a direct isometry given by

$$R_{\theta,P} = T_{\mathbf{p}} \circ U_\theta \circ T_{-\mathbf{p}}.$$

Theorem. An affinity T of the plane of the form $T(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$, is a rotation if and only if A is an orthogonal matrix with determinant 1. If the rotation is not trivial ($A \neq I$) it has a single fixed point with positive vector $\mathbf{p} = (I - A)^{-1}\mathbf{b}$, and the angle of the rotation is θ where $A = U_\theta$ by the rotation definition.

Definition. (Displacement) An isometry is called a displacement if it is either a rotation or a translation.

Theorem. A displacement that has no fixed points is a non-trivial translation; rotations fix exactly one point or all points. Furthermore:

i) The composition of two rotations is either a rotation or a translation.

ii) The composition of a rotation and a translation is a rotation through the same angle.

iii) The set of all displacement is a group under composition.

Theorem. Given two pairs of points A, B and A', B' with $d(A, B) = d(A', B')$ there is exactly one direct isometry T such that $T(A) = A'$ and $T(B) = B'$, and it is a displacement.

Corollary. Every direct isometry is a displacement.

3.3.2 Indirect Isometries

Definition. (Reflection) The transformation R_ℓ is a reflection in the line ℓ if for all P, if $P' = R_\ell(P)$, PP' is orthogonal to ℓ and $d(P, M) = d(P', M)$ where $M = PP' \cap \ell$.

Theorem. Let R_ℓ be a reflection in a line ℓ. Then

i) $R_\ell \circ R_\ell = R_\ell^2$ is the identity.

ii) $R_\ell(P) = P$ if and only if $P \in \ell$.

iii) R_ℓ is an indirect isometry.

iv) If λ is a line perpendicular to ℓ then $R_\ell(\lambda) = \lambda$.

Theorem. Let R_ℓ and R_λ be reflections in the lines ℓ and λ respectively.

i) $R_\lambda \circ R_\ell$ is a displacement.

ii) If ℓ and λ are parallel then $R_\lambda \circ R_\ell = T_{2\mathbf{a}}$, where $\mathbf{a}$ is the vector displacement from ℓ to λ.

iii) If $X = \ell \cap \lambda$ exists and θ is the (signed) angle from ℓ to λ then $R_\lambda \circ R_\ell = R_{2\theta,X}$.

Lemma. Let A be a real orthogonal 2×2 matrix with determinant -1. Then $A^2 = I$, A has eigenvalues 1 and -1 and so is diagonalisable.

Theorem. An affinity T on $\mathbb{R}^2$ of the form $T(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$, is a reflection if and only if A is an orthogonal matrix with determinant -1 and $A\mathbf{b} = -\mathbf{b}$. The line of reflection has parametric vector form $\mathbf{x} = \lambda\mathbf{u} + \frac{1}{2}\mathbf{b}$ where $\mathbf{u} \cdot \mathbf{b} = 0$.

Proposition. Any rotation can be written as the product of two reflections.

Definition. (Glide Reflection) An isometry formed from a reflection followed by an arbitrary translation is called a glide reflection.

Theorem. Given two pairs of points A, B and A', B' with $d(A, B) = d(A', B')$ there is exactly one indirect isometry T such that $T(A) = A$ and $T(B) = B'$, and it is a glide reflection.

Corollary. Every indirect isometry is either a reflection or a glide reflection.

3.3.3 Summary and a Criterion

To summarise, any isometry must be of the form $T(\mathbf{x}) = A\mathbf{x} + \mathbf{a}$ with A orthogonal ($A^T A = I$) and $\mathbf{a}$ fixed (possible zero).

Name	Type	$\det(A)$	Fixed points/lines	Invariant points/lines
translation by $\mathbf{b}$	direct	1	none	lines parallel to $\mathbf{b}$
rotation around P	direct	1	P	none (unless half or full turn)
reflection in ℓ	indirect	-1	ℓ	lines orthogonal to ℓ
glide reflection	indirect	-1	none	one line

Theorem. An isometry is determined by its effect on three non-collinear points.

3.4 Similarities

A similarity is a transformation that preserves shape but no necessarily size.

Definition. (Similarity) A transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ is a similarity if there is a constant $k > 0$ such that $d(T(A), T(B)) = kd(A, B)$ for all $A, B \in \mathbb{R}^2$. The constant k is the same for every A and B, and is known as the ratio of T.

If we are paying heed to the direction of lines, then we would allow $k < 0$ too. A similarity scales all lengths in all directions the same: it is an **isotropic scaling**. Similarities with $k > 0$ are also called **dilatations** or **dilations**. If the origin is preserved a similarity is sometimes called a **homothety**.

Definition. (Similar) If F_1 and F_2 are geometric objects related by a similarity we say they are similar.

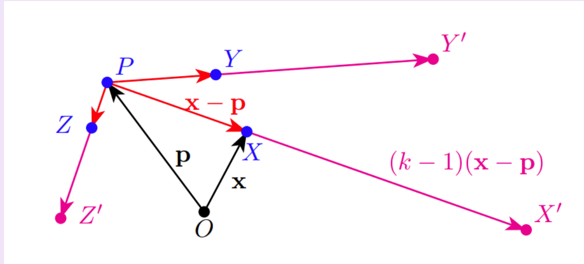
Theorem. The set of Similarities under composition is a subgroup of $\mathcal{A}_2$ and denoted $\mathcal{SM}_2$. Also the set of isometries, $\mathcal{I}_2$ is a subgroup of $\mathcal{SM}_2$.

Theorem. A similarity S preserves betweenness, concurrence, collinearity, line segments, triangles, polygons, (sizes of) angles, circles, ellipses (or any conic section), parallelism and ratios of distances.

Theorem. A similarity is determined by its effect on three non-collinear points.

Definition. (Radial Transform) A radial transform with centre P and ratio $k \neq 0$ is the map $D_{k,P}$ given by

$$D_{k,P}(\mathbf{x}) = k(\mathbf{x} - \mathbf{p}) + \mathbf{p} = k\mathbf{x} + (1 - k)\mathbf{p}.$$



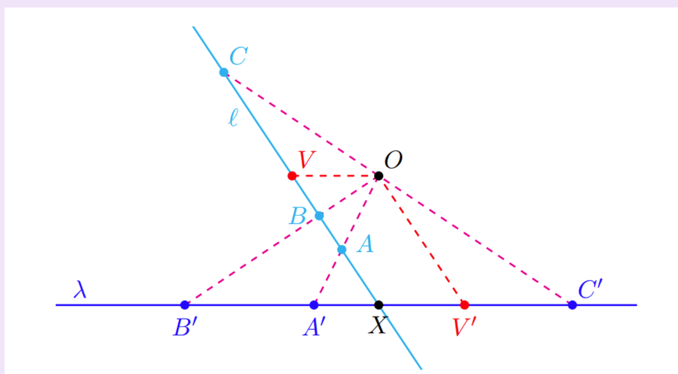
Proposition. For a fixed point P the set of radial transforms with centre P are an abelian subgroup of $\mathcal{SM}_2$.

Theorem. Every similarity, S, of ratio k is of the form $D_{k,P} \circ T$ for some isometry T and some point P.

4 Projective Geometry

4.1 Perspectivities of Lines

Definition. (Perspective Transformation) Let ℓ and λ be distinct lines and O a fixed point on neither. A perspective transformation of the lines, also called a perspectivity or central projection is a map $p : \ell \to \lambda$ that takes every point $P \in \ell$ to the point where OP meets λ, if such a point eixsts.



As the points on ℓ approach V, their images on λ get further and further away to the left until at V, when OV is parallel to λ, the image point vanishes: we call V the **vanishing point** on ℓ.

Definition. (Projectively Extended Real Line) The projectively extended real line, $\overline{\mathbb{R}}$ is the set $\mathbb{R} \cup \{\infty\}$ where ∞ is the point at infinity.

Definition. (Projective Transformation) A projective transformation or projectivity is any composition of perspectivities.

Any projectivity on a line that has more than two fixed points must be the identity. It also follows that a projectivity of a line to itself is uniquely defined given the images of three distinct points.

Theorem. Suppose $\{A, B, C, D\}$ and $\{A', B', C', D'\}$ are two sets of four coolinear points (not necessarily in the same plane) such that $[A, B; C, D] = [A', B'; C', D']$. Then there is a projectivity that maps A to A' etc.

4.2 Perspectivities of Planes

Definition. (Perspective Transformation) A perspective transformation, perspectivity or central projection between planes Π and $\Pi'$from $O \notin \Pi \cup \Pi'$ is given by mapping a point P on Π to a point P' on Π', if it exists, where O, P and P' are collinear.

Theorem. Let Π have Cartesian coordinates (x, y) and Π' Cartesian coordinates (x', y'). Then a perspectivity from a point O on neither plane maps coordinates by

$$x' = \frac{a_1 x + b_1 y + c_1}{a_3 x + b_3 y + c_3}, \quad y' = \frac{a_2 x + b_2 y + c_2}{a_3 x + b_3 y + c_3},$$

where the constants a_i, b_i etc satisfy $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \neq 0$. If the plans are parallel, then $a_3 = b_3 = 0$ and the perspectivity is an affinity.

Definition. (Projectivity) A composition of perspectivities of planes is called a projectivity of planes.

Theorem. Let T be a perspectivity of planes. THen T preserves collinearity and rectilinear figures. For lines and figures not crossing the vanishing line, T preserves betweenness and line segments, and hence polygons. In general T does not preserve betweenness, parallelism, line segments, distance and ratios of distance.

Theorem. (a) Every conic is mapped to another conic under a perspectivity.

(b) A degenerate conic is mapped to a degenerate conic under a perspectivity.

(c) A non-degenerate conic is mapped to a non-degenerate conic under a perspectivity and is the image of a circle under some perspectivity.

Since collinearity and coincidence are preserved under perspectivities, we can now generalise Pascal's Theorem to any non-degenerate conic $\mathcal{C}$.

4.3 The Real Projective Plane

Definition. (Real Projective Plane) The real projective plane, $\mathbb{RP}^2$, is the set whose elements are pairs of antipodal points on the (2-dimensional) sphere S^2. The lines in $\mathbb{RP}^2$ are the great circles on the sphere, with the antipodal points identified.

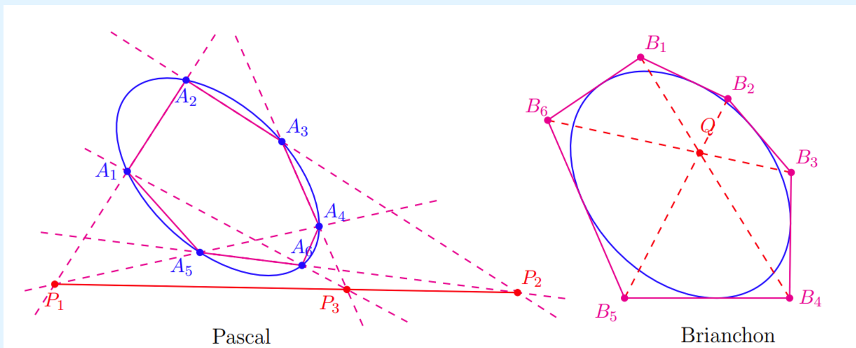
You should think of $\mathbb{RP}^2$ as what we get from identifying the image points on S^2 with the plane $\mathbb{R}^2$ under the gnomonic projection, but with the addition of an ideal line "at infinity" corresponding to the equator. In other words, $\mathbb{RP}^2$ is $\mathbb{R}^2 \cup \ell_\infty \cup \{\infty\} = \mathbb{R}^2 \cup \overline{\mathbb{R}}$, where we have added an extended real line to the plane: this is our second model of $\mathbb{RP}^2$.

As we know, in standard Euclidean geometry, any pair of points defines a unique line, and the same is still true in $\mathbb{RP}^2$. But in $\mathbb{RP}^2$ we now have any pair of lines defining a unique points, their intersection.

In fact, in $\mathbb{RP}^2$ (either model) we have the **principal of plane duality**: any theorem (or definition) involving lines and points remains true (or is equivalent) if we interchange the words "point" and "line". We also need to exchange "concurrent" and "collinear" of course, and we not that the dual of a point on a curve is the tangent at the at point. We sometimes use the word **join** for both concepts dually: the join of two points is the line going through them both and the join of two lines is the point where they meet.

The dual of Pascal's Theorem is the following:

Theorem. **(Brianchon's Theorem)** Suppose hexagon with vertices $B_1, \ldots, B_6$ has edges B_1B_2, B_2, B_3 etc tangent to a non-degenerate conic section. Then the lines B_1B_4, B_2B_5 and B_3B_6 joining opposite veritces are concurrent.



4.3.1 Synthetic Geometry in $\mathbb{RP}^2$

Taking the $\mathbb{R}^2 \cup \ell_\infty$ model of $\mathbb{RP}^2$, we consider the ratio $\frac{AC}{CB}$ where exactly one of the points is ideal:

1. if A is ideal, then we are dividing an infinite length by a finite one, so $\frac{A_\infty C}{CB} = \infty$;
2. if B is ideal, then we are dividing a finite length by an infinite one, so $\frac{AC}{CB_\infty} = 0$;
3. if C is ideal, then we are dividing an infinite length by another infinite one, as their direction are opposite we adopt the useful convention that $\frac{AC_\infty}{C_\infty B} = -1$.

Theorem. **(Menelaus' Theorem in $\mathbb{RP}^2$)** Let $\triangle ABC$ have at most one ideal vertex, and let D, E and F be non-vertex points on sides BC, CA, AB (or their extensions, and possibly ideal) respectively. Then D, E and F are collinear if and only if

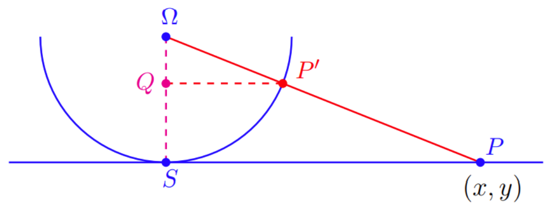
$$\frac{AF}{FB} \cdot \frac{BD}{DC} \cdot \frac{CE}{EA} = -1.$$

Definition. **(Perspective from points and lines)** We say that $\triangle ABC$ and $\triangle A'B'C'$ are perspective from a point P if the lines AA', BB' and CC' are concurrent at P. We say that $\triangle ABC$ and $\triangle A'B'C'$ are perspective from a line ℓ if the points $AB \cap A'B', BC \cap B'C'$ and $CA \cap C'A'$ all lie on ℓ.

Theorem. **(Desargues' Theorem (1648))** Two real triangles $\triangle ABC$ and $\triangle A'B'C'$ are perspective from some point if and only if they are perspective from some line.

4.3.2 Conic Sections on $\mathbb{RP}^2$

To help study conics on $\mathbb{RP}^2$ we need a formula for gnomonic projection:



As $|\Omega S| = 1$ and $\triangle \Omega P S \approx \triangle P'Q$, so $\Omega Q = (1 + x^2 + y^2)^{-1/2}$.

$$\overrightarrow{\Omega P} = (x, y, -1) \text{ so } \overrightarrow{\Omega P'} = \frac{1}{(1 + x^2 + y^2)^{1/2}}(x, y, -1).$$

So P' has coordinates $(0, 0, 1) + \frac{1}{(1+x^2+y^2)}^{1/2}(x, y, -1)$. The equivalent point in the upper hemisphere is

$$(0, 0, 1) + \frac{1}{(1 + x^2 + y^2)^{1/2}}(-x, -y, 1).$$

4.3.3 Third Model of $\mathbb{RP}^2$

Clearly every ray through the origin in $\mathbb{R}^3$ intersects the unit sphere S^2, in two antipodal points. So the set of such rays, $\mathcal{R}(\mathbb{R}^3)$, is in one-to-one correspondence with the points of $\mathbb{RP}^2$ in our original model of the sphere under gnomonic projection.

This is our third model of $\mathbb{RP}^2 : \mathcal{R}(\mathbb{R}^3)$, the set of all rays through the origin in $\mathbb{R}^3$.

We can use this correspondence to parametrise $\mathbb{RP}^2$. We can write any ray through the origin as $\{\lambda(a, b, c), \lambda \in \mathbb{R}\}$, where $(a, b, c) \neq \mathbf{0}$. We now define a map $T : \mathbb{RP}^2 \to \mathcal{R}(\mathbb{R}^3)$.

Firstly, let $(x, y) \in \mathbb{R}^2$. Then

$$\begin{aligned} T(x, y) &= \{\lambda(x, y, 1), \lambda \in \mathbb{R}\} \\ &= \{(X, Y, Z) : x = X/Z, y = Y/Z, Z \neq 0\} \in \mathcal{R}(\mathbb{R}^3). \end{aligned}$$

Secondly, if $m \in \ell_\infty$, then

$$T(m) = \{\lambda(1, m, 0), \lambda \in \mathbb{R}\} = \{(X, Y, 0) : m = Y/X, X \neq 0\} \in \mathcal{R}(\mathbb{R}^3).$$

This leaves the point ∞, so we define

$$T(\infty) = \{\lambda(0, 1, 0), \lambda \in \mathbb{R}\} = \{(0, Y, 0) : Y \neq 0\} \in \mathcal{R}(\mathbb{R}^3).$$

Definition. (Homogenous Coordinates) We call (X, Y, Z) the homogenous coordinates of the ray through the origin, and by extension the homogenous coordinates of a point in $\mathbb{RP}^2$.

4.4 Analytic Projective Geometry

4.4.1 Projectivities

A projective transformation between two planes

$$x' = \frac{a_1x + b_1y + c_1}{a_1x + b_3y + c_3}, \quad y' = \frac{a_2x + b_2y + c_2}{a_3x + b_3y + c_3},$$

where the constants a_1, b_i etc satisfy $\Delta = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \neq 0$, in homogenous coordinates is the more symmetric

$$X' = a_1X + b_1Y + c_1Z, \quad Y' = a_2X + b_2Y + c_2Z, \quad Z' = a_3X + b_3Y + c_3Z$$

with the same restriction on Δ.

Theorem. A projectivity is an affinity if and only if it leaves the ideal line invariant. It follows that $\mathcal{A}_2$ is a subgroup of $\mathcal{P}(\mathbb{RP}^2)$.

Theorem. For any two sets of four points $\{A, B, C, D\}$ and $\{A', B', C', D'\}$ in $\mathbb{RP}^2$, where no three of $\{A, B, C, D\}$ and no three of $\{A', B', C', D'\}$ are collinear, there is a unique projectivity T such that $T(A) = A', T(B) = B', T(C) = C'$ and $T(D) = D'$.

It follows that all quadrilaterals are projectively equivalent.

4.4.2 Projective Lines

Theorem. Points satisfying a homogenous equation of degree 1

$$aX + bY + cZ = 0,$$

where a, b and c are not all zero, form a line in $\mathbb{RP}^2$, and conversely all lines in $\mathbb{RP}^2$ have such an equation. If $a = b = 0$ this is a line at infinity. Otherwise the line contains exactly one ideal point with coordinates $-b, a, 0$.

4.4.3 Conic Sections Again

Theorem. If quadratic form

$$aX^2 + bY^2 + cZ^2 + 2dXY + 2eXZ + 2fYZ = 0,$$

is a non-degenerate conic section, then it is an ellipse, parabola or hyperbola if it has no, one or two ideal points respectively.

For a parabola, the ideal point is the intersection of ℓ_∞ and a line parallel to the axis of the parabola.

For a hyperbola, the ideal points are the intersections of ℓ_∞ with lines parallel to the asymptotes.

5 Group

We introduced the notion of groups in Section 3.1 and we will build on this idea and explore further geometrical applications.

Definition. (Order of Group) For a gorup G with a finite number of elements, the order of the group is the number of elements in G. We write this as $|G|$.

Theorem. Let G be a group. Then

a) The identity element is unique.

b) For any $g \in G$, the inverse g^{-1} is unique.

c) We can cancel elements if $ab = ac$ then $b = c$.

5.1 Small Finite Groups

Definition. (Cyclic Group) If n is a positive integer we can define an abstract group of order n called the cyclic group, C_n, by

$$C_n = \langle a \mid a^n = e \rangle = \{e, a, a^2, \ldots, a^{n-1}\},$$

with $a^n = e$.

Proposition. All cyclic groups are abelian.

5.2 Group Morphisms

Definition. (Isomorphism) A map $\phi : G \to H$ between two groups is an isomorphism if it is a bijection such that for all $g_1, g_2 \in G$,

$$\phi(g_1 \cdot g_2) = \phi(g_1) * \phi(g_2).$$

The groups G and H are then isomorphic.

Proposition. If G and H are finite and isomorphic, then $|G| = |H|$.

Definition. (Homomorphism) A map $\phi : G \to H$ between two groups is a homomorphism if for all $g_1, g_2 \in G$

$$\phi(g_1 \cdot g_2) = \phi(g_1) * \phi(g_2).$$

Theorem. Let G and H be groups and $\phi : G \to H$ a homomorphism. Then

a) $\phi(e_G) = e_H$

b) For all $g \in G, \phi(g^{-1}) = (\phi(g))^{-1}$.

c) For all $g \in G, \phi(g^n) = (\phi(g))^n$ for all $n \in \mathbb{Z}$.

d) The image of $\phi, \mathrm{im}(\phi)$, is a subgroup of H, and $\mathrm{im}(\phi)$ is isomorphic to G if ϕ is one-to-one.

5.3 The Symmetric and Dihedral Groups

Definition. (Permutation) A permutation on a finite set, S, of n distinct objects is a bijection $\sigma : S \to S$.

Theorem. The set of all permutations on $\{1, 2, 3, \ldots, n\}$ is a group of order $n!$.

Definition. (Symmetric Group) We call the group of all permtuations on n objects the symmetric group (on n letters), denoted $\mathcal{S}_n$.

Definition. (Dihedral Group) The symmetry group of the regular n-gon is the group of $2n$ elements D_{2n}, known as the dihedral group.

Theorem. Suppose G is a finite subgroup of linear isometries of the plane. Then G is isomorphic to C_n or C_{2n} for some positive integer n.

5.4 Order and Subgroups

Definition. (Order of Element) For a group G, and an element $g \in G$, the order of $g, \mathrm{ord}(g)$, is the smallest positive integer n for which $g^n = e$, if such an integer exists.

Proposition. For any group G and $g \in G$ of order n, the set $H = \{e, g, g^2, \ldots, g^{n-1}\}$ is a (cyclic) subgroup of G of order n.

Definition. (Generators) We say that H in the previous result is generated by g, and usually write the group generated by all powers of an element as $\langle g \rangle$.

Theorem. (Lagrange's Theorem) If G is a finite group and H is a subgroup of G, then $|H|$ is a factor of $|G|$.

Definition. (Cosets) Let G be a group and H a subgroup of G. For any $g \in G$ we define the left coset of H by g to be

$$gH = \{gh \mid h \in H\}.$$

There is a corresponding definition of a right coset of H by g to be

$$Hg = \{hg \mid h \in H\}.$$

Theorem. Let g be an element of a group G. Then

a) the order of g is equal to the order of the subgroup $\langle g \rangle$;

b) the order of g is a factor of the order of the group $|G|$;

Theorem. The only group of prime order are the cyclic groups, C_p, and their only subgroups are the trivial ones, $\{e\}$ and C_p.

Corollary. All groups of prime order are abelian.

5.5 Normal subgroups and Quotient Groups

Definition. (Normal) A subgroup H of group G is normal if for all $g \in G, gH = Hg$, or equivalently, $gHg^{-1} = H$.

Lemma. Let H be a subgroup of G. If for every $g \in G$ and $h \in H$ the element $g.h.g^{-1}$ is in H, then H is normal.

Proposition. Suppose G is a group of order $2k$. Then any subgroup of order k is normal.

Theorem. Let G be a subgroup of the isometries $\mathcal{I}_\in$ and N the subset of translations in G. Then N is a normal subgroup of G.

Theorem. Let G be any group and N a normal subgroup. Then the set of all cosets, $\{xN\}$, is a group under the operation

$$(xN)(yN) = (xy)N.$$

If $|G|$ is finite the order of this group is $|G|/|N|$.

Definition. (Quotient Groups) The group of cosets in the above theorem is called the factor group or quotient group, and denoted G/N.

Theorem. If N is a normal subgroup of G then the map $\phi : G \to G/N$ defined by $\phi(g) = gN$ is a homomorphism.

Definition. (Natural Homomorphism) The map in the above theorem is called the natural homomorphism from G to G/N.

Definition. (Kernel) Let $\phi : G \to H$ be a homomorphism of groups. The kernel of $\phi, \ker\phi$, is the set of all elements that maps to the identity (in H):

$$\ker(\phi) = \{g \in H \mid \phi(g) = e_H\}.$$

Proposition. A homomorphism $\phi : G \to H$ is one-to-one (injective) if and only if $\ker(\phi) = \{e_G\}$.

Theorem. The kernel of any homomorphism $\phi : G \to H$ is a normal subgroup of G.

Theorem. (First Isomorphism Theorem) For any homomorphism $\phi : G \to H$, the factor group $G/\ker(\phi)$ is isomorphic to $\mathrm{im}(\phi)$.

5.5.1 Group Theory vs Linear Algebra

Recall that two vector spaces over the same field are isomorphic (as vector spaces) if and only if they have the same dimension. Thus if we are translating group theory to linear algebra, we can replace isomorphic with have the same dimension.

In fact, the first isomorphism theorem is just the group theory version of linear algebra's rank-nullity theorem.

5.6 Product Groups

Definition. (Product Groups) Let $(G, \cdot)$ and $(H, *)$ be two groups. The (direct) product group, is the set of ordered pairs

$$G \times H = \{(g, h) \mid g \in G, h \in H\}$$

with group operation

$$(g_1, h_1)(g_2, h_2) = (g_1 \cdot g_2, h_1 * h_2).$$

If G and H are finite then clearly $|G \times H| = |G| \cdot |H|$.

Proposition. Let G and H be two groups. Then

a) $G \times H$ and $H \times G$ are isomorphic.

b) If G and H are abelian, so is $G \times H$.

5.6.1 Groups of Order 8

So far we have

1. C_8 — cyclic (and hence abelian)
2. D_8 — the symmetry group of the square; not abelian.
3. $C_2 \times C_2 \times C_2$ — abelian.

One more obvious group of order 8 is $C_4 \times C_2$, also abelian. There is in fact a fifth group of order 8, Q_8, the quaternion group. Its most basic representation is the linear one on $\mathbb{C}^2$.

$$Q_8 = \left\{ \pm \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \pm \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \pm \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \pm \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} \right\}$$

But we can more compactly represent the group on the quaternions.

Definition. (Quaternions) The quaternions, $\mathbb{H}$, are the entities formed by taking real linear combinations of the four basis elements $\{1, i, j, k\}$ where $i^2 = j^2 = k^2 = ijk = -1$.

A quaternion looks like $a = a_0 + a_i i + a_j j + a_k k$ for real a_0, a_i, a_j, a_k. Multiplication of general quaternions is defined in the obvious way, but it is not commutative.

We can now represent Q_8 as the group pf unit quaternions,

$$\{\pm 1, \pm i, \pm j, \pm k\}$$

under multiplication in $\mathbb{H}$.

5.6.2 Symmetrices of the Cube

The physical symmetry group of the cube has order 24 (it would be 48 if we allowed reflections).

a) One identity element, of order 1.

b) 9 face-centred rotations: i.e. through $\pm\frac{1}{2}\pi$ or π about lines through the centres of the three paris of opposite faces.

c) 8 vertex-centred rotations: i.e. through $\pm 2\pi/3$ around the four space diagonals.

d) 6 edge-centred rotations: i.e. through π about lines joining midpoints of the six paris of opposite edges.

This group turns out to be isomorphic to $\mathcal{S}_4$: the group of permutation of 4 objects.

5.7 Frieze Groups

We have so far met two principal types of symmetry groups

a) finite ones, all realizable as the symmetry group of some geometric object;

b) infinite ones depending on some continuous parameter, such as rotations about the origin. The symmetry group of a circle is an obvious example.

The finite symmetry groups allow rotations through a discrete set of angles only (e.g. $\frac{1}{2}\pi$ for a square).

There are in fact, patterns whose symmetry groups are infinite but consist of discrete isometries. The simplest sort are the **frieze groups**, where there is a 2-dimensional pattern repeating on one dimension (a **frieze**). Such groups have translations in one direction only, and to be discrete, the translations must all be integer multiples of some minimal translation.

5.8 Wallpaper Group

Wallpaper groups are the equivalent to frieze groups btu with two independent minimal translations.

Definition. (Orbit and Stabliziers) Let Gb e a group of transformations in $\mathbb{R}^2$ and $P \in \mathbb{R}^2$ be any point.

The **orbit** of P under $G, \mathcal{O}(P)$, is the set of all points of $\mathbb{R}^2$ you get from applying an arbitrary member of G to P. The **stabilizer** of P, G_P, is the set of elements of G that fix P.

Clearly for any two points X and Y in the orbit of P there is an element of G taking X to Y. It is also not hard to see that the stabilizer of a point is a subgroup of G and the same (isomorphic) at every point of a given orbit. It is also called the **isotropy group** of a point (or orbit).

Theorem. For a wallpaper group G with translation subgroup $N, G/N$ is isomorphic to G_O, the stabilizer of the origin.

Theorem. For a wallpaper group G, the isotropy group of the origin, G_O is C_n or D_{2n} for some integer n.

Theorem. (The Crystallographic Restriction Theorem) Let the symmetry group, G, of a 2-dimensionallattice admit both translations and rotations about one of its lattic points. Then those rotations can only be of period $1, 2, 3, 4$ or 6 i.e. through multiples of $2\pi/n$ for $n = 1, 2, 3, 4$ or 6.

5.9 Tilings

Closely related to the idea of wallpaper groups is the concept of a tessellation or tiling of the plane: covering the plane completely with copies of one (or more) geometric shapes ("tiles").

Lemma. The composition of two half turns around point P and then Q is a translation through $2\overline{PQ}$.